CHEYANNE SHARIAT

cshariat@caltech.edu ◇ cheyanneshariat.github.io

EDUCATION

Ph.D., Astrophysics, California Institute of Technology — 2024 -
M.S., Astrophysics, California Institute of Technology — 2024 - 2026
DOE Computational Science Graduate Fellow
Advisor: Kareem El-Badry

B.S., Physics, University of California, Los Angeles — 2021 - 2024
summa cum laude
Tisch Academic Excellence Scholar
Advisor: Smadar Naoz

RESEARCH POSITIONS

Graduate Student Researcher — 2024 -
California Institute of Technology — *Pasadena, CA*

Undergraduate Student Researcher — 2021 - 2024
University of California, Los Angeles — *Los Angeles, CA*

Summer Undergraduate Research Fellow — 2023
NASA Jet Propulsion Laboratory (JPL) — *Pasadena, CA*

Summer Undergraduate Research Fellow — 2022
Caltech High Energy Astrophysics Group — *Pasadena, CA*

High School Researcher — 2019 - 2022
UCLA Neurology and Neurogenetics, Fogel Lab — *Los Angeles, CA*

GRANTS

Roman General Investigator Grant, NASA, PI, $450,000 — 2027 – 2030
Binary Star Demographics in the Galactic Bulge
Program ID: 19046

Roman General Investigator Grant, NASA, PI, $450,000 — 2027 – 2030
RACS: Roman Astrometric Companion Survey
Program ID: 19028

Roman General Investigator Grant, NASA, Co-I $450,000 — 2027 – 2030
RAGHAB: Roman Analysis for a Galactic Hunt for Astrometric Binaries
Program ID: 19080

JWST General Observer Grant, NASA/STScI, PI, $205,225 — 2026 – 2029
Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87
Program ID: 11051

HONORS & AWARDS

DOE Computational Science Graduate Fellowship — 2025 – 2029

NSF Graduate Research Fellowship (offer declined) — 2025

Raynor L. Duncombe Student Research Prize, AAS DDA — 2025

Joshua and Beth Friedman Fellowship, Caltech — 2025

Dean's Prize for Excellence in Research, UCLA — 2024

Lindau Nobel Laureate Meetings Fellow, Lindau, Germany	2024
Helen Quinn Award for Undergraduate Research Theory (1st), APS	2023
Summer Undergraduate Research Fellowship, NASA Jet Propulsion Laboratory	2023
Undergraduate Research Fellowship, UCLA	2022 – 2023
Summer Undergraduate Research Fellowship, Caltech	2022
Steve Tisch Academic Excellence Scholarship, UCLA	2021 – 2024

TALKS

Explore JPL, Talk, NASA/JPL, Pasadena, CA	2026
Lunch Talk, University of California, Berkeley, CA	2026
Galform, Talk, University of California, Berkeley, CA	2026
EuroWD, Talk, Vienna, Austria	2026
Galaxy Coffee, Talk, MPIA Heidelberg, Germany	2026
DOE Annual Program Review, Poster, Washington D.C.	2026
Stellar Thinkshop, Talk, Carnegie Observatories, Pasadena, CA	2026
IAS Astro Coffee, Talk, Princeton, NJ	2026
Thursday Lunch Talk, Princeton, Princeton, NJ	2026
Stellar-Mass Black Holes Workshop, Talk, KITP	2025
The Lifecycle of Stellar Black Holes Conference, Talk, KITP	2025
Carnegie ArXiv Tea, Pasadena, CA	2025
Keck Science Meeting, Poster, UCLA	2025
Binary Stars in the Space Era, Keele, United Kingdom	2025
Vasto Accretion Meeting, Vasto, Italy	2025
AAS Division on Dynamical Astronomy Meeting 56, Atlanta, GA	2025
Undergraduate Research Week, UCLA	2024
Exoplanet Monthly Meeting, Northwestern, Evanston, IL	2024
APS Conference for Undergraduate Women in Physics, San Diego, CA	2024
APS Far West Section, UCSD (Helen Quinn Award, 1st)	2023
Division 326 Seminar, NASA/JPL, Pasadena, CA	2023
Undergraduate Research Week, UCLA	2023
SURF Final Presentation, Caltech	2022
HEAG Presentation Week, Caltech	2022

AWARDED TELESCOPE TIME

James Webb Space Telescope	24.8 hours (PI)
Palomar Hale 200 inch, NGPS	8 nights (PI) 5 nights (Co-I)
Palomar Hale 200 inch, PHARO	9 nights (PI)
Keck HIRESb	9 nights (Co-I)

Keck KCWI 1 night (Co-I)

Swift UVOT, TOO 3000 s (PI)

OBSERVING EXPERIENCE

Palomar Hale Telescope, PHARO – 10 nights 2024-

Keck, HIRES – 3 nights 2025-

MENTORING

Joe Smith (Caltech Undergrad) 2025-

Noah East (UCLA Undergrad) 2025-

Omri Ratzkoff (UCLA Undergrad) 2025-

JOURNAL REFEREE

ApJ, ApJL, MNRAS, A&A, A&AL (*15+ papers*) 2023 -

COLLABORATIONS

Sloan Digital Sky Survey 2026 -

LISA Consortium 2025 -

Euclid Consortium 2025 -

TEACHING & OUTREACH

Graduate Teaching Assistant (Caltech) 2025
Ay 101: Physics of Stellar Interiors and Atmospheres

Graduate Peer Mentor (Caltech) 2025 -

LAUSD Guest Speaker 2023 -
James Monroe High School
Shenendoah Early Education Center

Stargazing Lectures (Caltech) 2023 -
Q&A Panelist
Telescope Operator

Diversity, Equity, and Inclusion (DEI) Committee 2023 - 2024
Undergraduate Representative, UCLA Physics and Astronomy Dept. *Los Angeles, CA*

Undergraduate Learning Assistant 2022
Physics 1C, Electrodynamics, Optics, and Special Relativity, UCLA *Los Angeles, CA*

PUBLICATIONS (26 TOTAL; 16 FIRST-AUTHOR)

26. **Shariat, C.**, El-Badry, K., and Naoz, S., "A concordance model of neutron star kicks", arXiv:2610.04005, 2026.

25. Ding, K., Gennaro, M., Avila, R. J., Ricotti, M., Beaton, R. L., Boyer, M. L., Brown, T. M., Calamida, A., Cassisi, S., Chandra, V., Cohen, R. E., Correnti, M., Crnojević, D., El-Badry, K., Geha, M., Guhathakurta, P., Kallivayalil, N., Kirby, E. N., McQuinn, K. B. W., Savino, A., **Shariat, C.**, Simon, J. D., and Weisz, D. R., "Probing the Initial Mass Function in the Early Universe: Direct Measurements in the Boötes I Ultrafaint Dwarf Galaxy with JWST/NIRCam", ApJ, vol. 1009, Art. no. 94, arXiv:2605.15069, 2026.

24. **Shariat, C.**, Yamaguchi, N., El-Badry, K., and Maccarone, T. J., "Gaia neutron stars: demographics, birth rates, and connections to other neutron star populations", arXiv:2610.00454, 2026.

23. Yamaguchi, N., El-Badry, K., Essick, R., Farah, A., **Shariat, C.**, and Shahaf, S., "Beyond what's observed: hierarchical inference of Gaia astrometric compact-object binaries", arXiv:2609.35999, 2026.

22. Sanghi, A., Quarles, B., Lin, A. S. J., **Shariat, C.**, Kervella, P., Mang, J., Mawet, D., Giovinazzi, M. R., Gupta, A. F., Beichman, C., Mennesson, B., Mâlin, M., Xuan, J. W., Majumdar, L., Lissauer, J. J., Wyatt, M., Balmer, W. O., Bendek, E., and Wagner, K., "Worlds Next Door. V. A Candidate Solar System Scale Super-Jupiter in the 61 Cygni Binary System", arXiv:2609.32893, 2026.

21. **Shariat, C.**, El-Badry, K., Gull, M., Weisz, D. R., Conroy, C., and Savino, A., "A census of massive eclipsing binaries in the Whirlpool Galaxy", PASP, *submitted*, arXiv:2609.19268, 2026.

20. **Shariat, C.**, El-Badry, K., Wang, S., Wang, X.-Y., Rice, M., Xuan, J. W., and Ciardi, D. R., "Stellar companions sculpt hot Jupiter formation and spin-orbit evolution", PASP, *submitted*, arXiv:2609.19249, 2026.

19. Yamaguchi, N., El-Badry, K., and **Shariat, C.**, "Population Demographics of Post-interaction WDMS Binaries: From Common Envelope Evolution to Stable Mass Transfer", PASP, vol. 138, Art. no. 094209, arXiv:2607.12034, 2026.

18. Bhattacharjee, S., El-Badry, K., Fuller, J., **Shariat, C.**, and Yamaguchi, N., "Thermally Inflated Accretors in Post-mass Transfer Binaries: A35 and its Class Revisited", PASP, vol. 138, Art. no. 084202, arXiv:2603.23756, 2026.

17. **Shariat, C.**, El-Badry, K., MacLeod, M., and Leiner, E., "Gaia Astrometry Disfavors a Binary Origin for Long Secondary Periods", PASP, vol. 138, Art. no. 054202, arXiv:2604.09767, 2026.

16. **Shariat, C.**, El-Badry, K., and Bhattacharjee, S., "How precisely can we measure the ages of subgiant and giant stars?", OJAp, vol. 9, Art. no. 62534, arXiv:2510.08675, 2026.

15. **Shariat, C.**, "OverCite: Add Citations in LaTeX without Leaving the Editor", RNAAS, vol. 10, Art. no. 86, arXiv:2604.15366, 2026.

14. **Shariat, C.** and El-Badry, K., "A Global View of Post-interaction White Dwarf-main Sequence Binaries", PASP, vol. 138, Art. no. 034202, arXiv:2601.00439, 2026.

13. **Shariat, C.**, Ye, C. S., Naoz, S., and Rose, S. C., "Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-induced Channels", ApJL, vol. 1000, Art. no. L17, arXiv:2511.18678, 2026.

12. Holzknecht, L., Naoz, S., and **Shariat, C.**, "Possible Dynamical Pathways to the Misalignment of the VHS 1256–1257 System", ApJ, vol. 998, Art. no. 122, arXiv:2509.21452, 2026.

11. Xuan, Z., **Shariat, C.**, and Naoz, S., "From Wide Triples to Ultracompact X-ray Binaries: Multimessenger Signatures of Dynamically Formed Black Hole–White Dwarf Systems in the LISA Band", ApJ, vol. 995, Art. no. 27, arXiv:2508.13264, 2025.

10. **Shariat, C.**, El-Badry, K., Gennaro, M., Ding, K., Simon, J. D., Avila, R. J., Calamida, A., Cassisi, S., Correnti, M., Weisz, D. R., Geha, M., Kirby, E. N., Brown, T. M., Ricotti, M., McQuinn, K. B. W., Kallivayalil, N., Gilbert, K., Pacifici, C., Guhathakurta, P., Crnojević, D., Boyer, M. L., Beaton, R. L., Chandra, V., Cohen, R. E., Renzini, A., Savino, A., and Tollerud, E. J., "Wide Binaries in an Ultra-faint Dwarf Galaxy: Discovery, Population Modeling, and A Nail in the Coffin of Primordial black hole Dark Matter", PASP, vol. 137, Art. no. 104103, arXiv:2509.04555, 2025.

9. **Shariat, C.**, El-Badry, K., and Naoz, S., "10,000 Resolved Triples from Gaia: Empirical Constraints on Triple Star Populations", PASP, vol. 137, Art. no. 094201, arXiv:2506.16513, 2025.

8. **Shariat, C.**, El-Badry, K., Naoz, S., Rodriguez, A. C., and van Roestel, J., "Cataclysmic Variables in Triples: Formation Models and New Discoveries", PASP, vol. 137, Art. no. 074201, arXiv:2501.14025, 2025.

7. Matsuno, T., Kemp, A., Tanikawa, A., **Shariat, C.**, El-Badry, K. J., Dodd, E., Helmi, A., Koch-Hansen, A. J., Yamaguchi, N., and Yan, H., "Unevolved Li-rich stars at low metallicity: A possible formation pathway through novae", A&A, vol. 699, Art. no. A171, arXiv:2502.18552, 2025.

6. **Shariat, C.**, Naoz, S., El-Badry, K., Rocha, K. A., Kalogera, V., Stephan, A. P., Burdge, K. B., and Angelo, I., "Triple Evolution Pathways to Black Hole Low-mass X-Ray Binaries: Insights from V404 Cygni", ApJ, vol. 983, Art. no. 115, arXiv:2411.15644, 2025.

5. **Shariat, C.**, Naoz, S., El-Badry, K., Rodriguez, A. C., Hansen, B. M. S., Angelo, I., and Stephan, A. P., "Once a Triple, Not Always a Triple: The Evolution of Hierarchical Triples That Yield Merged Inner Binaries", ApJ, vol. 978, Art. no. 47, arXiv:2407.06257, 2025.

4. Stephan, A. P., Martin, D. V., Naoz, S., Hughes, N. R., and **Shariat, C.**, "Two Novel Hot Jupiter Formation Pathways: How White Dwarf Kicks Shape the Hot Jupiter Population", ApJL, vol. 977, Art. no. L11, arXiv:2408.13307, 2024.

3. **Shariat, C.**, Hasegawa, Y., Hansen, B. M. S., Yu, T. Y. M., and Hu, R., "Predicting the Dominant Formation Mechanism of Multiplanetary Systems", ApJL, vol. 964, Art. no. L13, arXiv:2402.04574, 2024.

2. **Shariat, C.**, Naoz, S., Hansen, B. M. S., Angelo, I., Michaely, E., and Stephan, A. P., "Dynamical Evolution of White Dwarfs in Triples in the Era of Gaia", ApJL, vol. 955, Art. no. L14, arXiv:2306.13130, 2023.

1. Lazzarini, M., Hinton, K., **Shariat, C.**, Williams, B. F., Garofali, K., Dalcanton, J. J., Durbin, M., Antoniou, V., Binder, B., Eracleous, M., Vulic, N., Yang, J., Wik, D., Gasca, A., and Kuauhtzin, Q., "Multiwavelength Characterization of the High-mass X-Ray Binary Population of M33", ApJ, vol. 952, Art. no. 114, arXiv:2305.16390, 2023.